1⃣ Abstract Super Class

Class: User (Abstract)

Attributes:

- userId : String
- name : String
- email : String
- password : String
- contactNo : String
- role : String
- accountStatus : String

Operations:

- login(email, password) : boolean
- logout() : void
- updateProfile(name, email, contactNo) : void
- changePassword(oldPwd, newPwd) : boolean
- viewDashboard() : void

2⃣ Specialized User Classes (Inheritance)

Class: Student (extends User)

Attributes:

- enrollmentNo : String
- batchId : String

Operations:

- viewAssignments() : List<Assignment>
- submitAssignment(assignmentId, file) : Submission
- reviseSubmission(submissionId, file) : void
- trackSubmissionStatus(assignmentId) : String
- viewEvaluation(submissionId) : Evaluation

Class: Faculty (extends User)

Attributes:

- facultyId : String
- department : String

Operations:

- createAssignment(details) : Assignment
- publishAssignment(assignmentId) : void
- unpublishAssignment(assignmentId) : void
- evaluateSubmission(submissionId, marks, remarks) : Evaluation
- extendDeadline(assignmentId, newDate) : void
- viewSubmissions(assignmentId) : List<Submission>
- generateReport(courseId) : Report

Class: Administrator (extends User)

Attributes:

- adminId : String

Operations:

- createUser(userDetails) : User
- updateUser(userId) : void
- deactivateUser(userId) : void
- createCourse(courseDetails) : Course
- updateCourse(courseId) : void
- createBatch(batchDetails) : Batch
- assignStudentToBatch(studentId, batchId) : void
- assignFaculty(courseId, batchId, facultyId) : void
- viewAuditLogs() : List<AuditLog>

3⃣ Core Academic Classes

Class: Course

Attributes:

- courseId : String
- courseCode : String
- courseName : String
- semester : int
- department : String

Operations:

- addAssignment(assignment) : void
- updateCourseDetails() : void
- getAssignments() : List<Assignment>

Class: Batch

Attributes:

- batchId : String
- batchName : String
- academicYear : String

Operations:

- addStudent(student) : void
- removeStudent(studentId) : void
- getStudentList() : List<Student>

Class: FacultyCourseBatchMapping

Attributes:

- mappingId : String
- facultyId : String
- courseId : String
- batchId : String

Operations:

- createMapping() : void
- removeMapping() : void

4⃣ Assignment Handling Classes

Class: Assignment

Attributes:

- assignmentId : String
- title : String
- description : String
- startDate : Date
- dueDate : Date
- maxMarks : int
- status : String (Published / Unpublished)

Operations:

- publish() : void
- unpublish() : void
- updateDetails() : void
- extendDeadline(newDate) : void
- getSubmissions() : List<Submission>

Class: Submission

Attributes:

- submissionId : String
- submissionDate : DateTime
- filePath : String
- status : String (Submitted / Late / Evaluated)
- studentId : String
- assignmentId : String

Operations:

- validateFile() : boolean
- storeSubmission() : void
- updateSubmission(file) : void
- updateStatus(status) : void

Class: Evaluation

Attributes:

- evaluationId : String
- marks : int
- remarks : String
- correctedFilePath : String
- evaluationDate : DateTime
- submissionId : String

Operations:

- assignMarks(marks) : void
- addRemarks(text) : void
- uploadCorrectedFile(file) : void
- saveEvaluation() : void

---

5. Supporting Classes

Class: Notification

Attributes:

- notificationId : String
- message : String
- date : DateTime
- userId : String
- status : String (Read/Unread)

Operations:

- sendNotification() : void
- markAsRead() : void
- getNotifications(userId) : List<Notification>

Class: Dashboard

Attributes:

- dashboardId : String
- userId : String

Operations:

- displayPendingAssignments() : void
- displaySubmissionStatus() : void
- displayEvaluationSummary() : void
- generateStatistics() : void

Class: AuditLog

Attributes:

- logId : String
- userId : String
- action : String
- timestamp : DateTime

Operations:

- recordAction() : void
- getLogs() : List<AuditLog>

🔗 RELATIONSHIPS BETWEEN CLASSES

1⃣ Inheritance (Generalization)

- User ⟶ Student
- User ⟶ Faculty
- User ⟶ Administrator

2⃣ Associations

- Course 1 —— * Assignment
- Assignment 1 —— * Submission
- Submission 1 —— 1 Evaluation
- Student 1 —— * Submission
- Faculty 1 —— * Assignment
- Batch 1 —— * Student
- FacultyCourseBatchMapping links Faculty + Course + Batch

- User 1 —— * Notification
- User 1 —— 1 Dashboard
- Administrator 1 —— * AuditLog

3⃣ Aggregation / Composition

- Assignment ◆—— Submission (Composition)
- Submission ◆—— Evaluation (Composition)
- Course ◇—— Assignment (Aggregation)

🎯 Diagram Drawing Tip for Lab

When drawing in UML:

- Use **abstract class (User)** at top
- Show inheritance arrows downward
- Use multiplicity (1, *, 0..1) clearly
- Show composition with filled diamond
- Show aggregation with hollow diamond